Exp: 13

AIM:

To implement a simple ping program using socket programming to check connectivity b/w source & destination hosts.

ALGORITHM:

1. Start the program
2. Get the destination IP address from user
3. Create a raw socket
4. Construct an ICMP echo request packet.
5. Send the packet to destination
6. Wait for an ICMP echo reply
7. If a reply is received, display Reply received.
8. If no reply is received within the timeout display request Timed out
9. Stop the program

CODE:

```
import socket
import struct
import time
host = input("Enter IP address")
s = socket.socket(socket.AF_inet, socket.
                  SOCK_RAW, socket.IPPROTO.ICMP)
s.settimeout(2)
packet = struct.pack("!BBHHH", 8,8,8,1,1)
s.sendto(packet, (host, 0))

try:
    data, addr = s.recvfrom(1024)
    print("Reply received from", addr[0])
except socket.timeout:
    print("Request timeout")
```

INPUT:

Enter destination IP address: 8.8.8.8

OUTPUT:

Ping 8.8.8.8
Reply received from 8.8.8.8

Result: Thus the ping program was successfully implemented using socket programming and the connectivity between the source and destination